



UNIVERSITY OF
MICHIGAN

EECS 453

Principles of Machine Learning

Lecture 1 – Introduction

Instructor: Prof. Qing Qu

Lecture Agenda

- Course Logistics & Prerequisites
- Homework, Exams & Grading
- Course Goals & Motivating Examples
- Course Content & Syllabus
- Books & References

Lecture Agenda

- **Course Logistics & Prerequisites**
- Homework, Exams & Grading
- Course Goals & Motivating Examples
- Course Content & Syllabus
- Books & References

Course Logistics

- **Lecture:** Mon./Wed., 12:00 PM-1:30 PM Eastern Time
- **Classroom Location:** EECS1303
- **Instructor:** Prof. [Qing Qu](#), email: qingqu@umich.edu,
- **Instructor office hour:** Wed. 1:30 PM – 3:00 PM (from Week 2)
- **Office hour location:** 4227 EECS

Course Logistics

- **Lecture:** Mon./Wed., 12:00 PM-1:30 PM Eastern Time
- **Location:** EECS1303
- **Teaching Format:** Hybrid Mode
 - The class will be taught in person, but will be live over Zoom simultaneously
 - In person is not mandatory but highly recommended
 - Each class will be recorded, and the videos will be released after class

Course Logistics

- **GSI:** Lianghe Shi, email: lhshi@umich.edu
- **Discussion Session:** Fri. 11:30 AM–12:30 PM, 165 CHRYS
- **GSI office hours:** Fri. 1:00 PM - 2:00 PM
- **Office hour location:** TBA

Course Logistics

- **Course materials:** Primarily posted on Canvas
- **Major References:**
 - [Foundations of Machine Learning](#), by Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar.
 - [Mathematics for Machine Learning](#), Marc Peter Deisenroth A. Aldo Faisal Cheng Soon Ong
 - [The Elements of Statistical Learning: Data Mining, Inference, and Prediction](#), by Trevor Hastie, Robert Tibshirani, and Jerome Friedman.
 - [Deep Learning](#), by Ian Goodfellow, Yoshua Bengio, and Aaron Courville.

Course Logistics

- **Piazza** will be the major place for Q&A.
- **No emails** of technical questions will be answered (Instructor & GSI), they will be addressed via **Piazza**.
- Personal emails (e.g., sickness) will be addressed.

Prerequisites

- **EECS 351:** Digital Signal Processing and Analysis
<https://ece.engin.umich.edu/academics/course-information/course-descriptions/eecs-351/>
- **EECS 301:** Probabilistic Methods in Engineering
<https://ece.engin.umich.edu/academics/course-information/course-descriptions/eecs-301/>
- **Basic Linear Algebra courses**
- **Basic Programming (in Python)**
EECS 280 <https://eecs280.org/>

Relationship to Other ML Courses

- **Related Courses:**

- EECS 445: entry-level ML course in CSE (undergraduate)
- EECS 545/553: more advanced ML in EECS (graduate)

- **This Course:**

- A new standard ML course for ECE undergraduate students
- Suitable for all students who want to learn basics of ML
- Does not have enough preparation for EECS 545/553
- Focus on basic principles while less programming, cover slightly different contents compared to EECS 445

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Homework, Exams & Grading

- Homework (5 assignments, 40%);
- Mid-term Exam (30%);
- Final Course Project (25%);
- Participation & Course Evaluation (5%).

More on Homework

- **Five** homework assignments (40%):
 - (Usually) release on **Friday**, due **two weeks after**;
 - Requests for due date extensions must be made in advance. Failure to do so will result in 50% assignment points off;
 - The maximum number of late requests is limited to **two**;
 - The maximum late time is limited to **three days**;
 - Please submit **.pdf** files online;
 - The one of the lowest grade will be dropped;

More on Homework

- **Five** homework assignments (40%):
 - **Regrading** request needs to be submitted **within 2 weeks** after grade released;
 - Homework assignments must be completed **on your own entirely**;
 - **Using existence of solutions** prepared in prior years is **not allowed**, violation of the policy is an Honor Code Violation;
 - Posting solutions and codes on **public sites** is also **prohibited**.

Study Group

- Form your study group early on!
- For homework, you may discuss between the study group members, but you should write your own solution **independently**. Never share your solutions!
- In the homework submissions, you must **put the names of your collaborators** if you have discussed ideas in depth
- Please start assignments early. (Warning: cramming does not work!)

Homework, Exams & Projects

- Homework (5 assignments, 40%);
- **Mid-term Exam (30%):**
 - Covers supervised learning and basic concepts;
 - No makeup exam, and **no final exam** 😊
 - We will perform one midterm review;
 - The mid-term will take place on **10/28/2026**.
- Final Course Project (25%);
- Participation & Course Evaluation (5%).

Homework, Exams & Grading

- Homework (5 assignments, 40%);
- Mid-term Exam (30%);
- **Final Course Project (25%);**
- Participation & Course Evaluation (5%).

Final Course Project

- **Apply machine learning to specific problems/dataset**
 - It should be the problem that you are interested;
 - You can form a group of maximum 2;
 - You can combine with your research if you are currently involved in some research projects.

Final Course Project

- **Apply machine learning to specific problems/dataset**
 - It should be the problem that you are interested;
 - You can form a group of maximum 2;
 - You can combine with your research if you are currently involved in some research projects.
- **Three Milestones:**
 - Project proposal (up to 1 page, due **11/08/2026**)
 - Final presentation (last week of the class, **12/09/2026**)
 - Final report (end of the semester, **12/18/2026**)
 - Further details about the project will be provided later in the quarter.

Homework, Exams & Grading

- Homework (5 assignments, 40%);
- Mid-term Exam (30%);
- Final Course Project (25%);
- **Participation & Course Evaluation (5%).**

Participation & Course Evaluation

- **Participation & Quiz (4%)**
 - It is based on the assessment of your engagement in the course;
 - We will have small quizzes during the class (from Week 2);
 - Each problem will be multiple choice question, worth 3 points.
 - Wrong answer: 2 points, no response: 0 points.

Participation & Course Evaluation

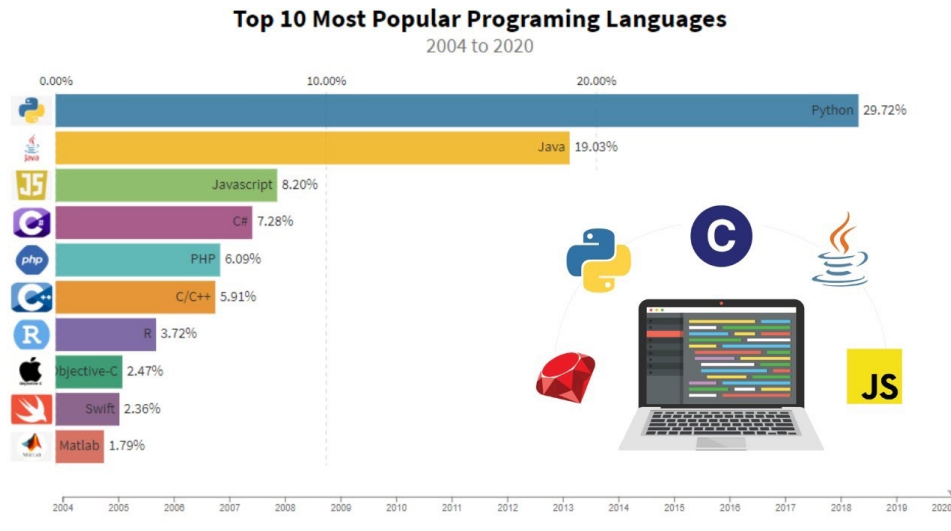
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 - We will have small quizzes during the class (from Week 2);
 - Each problem will be multiple choice question, worth 3 points.
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- **Course Evaluation (1%)**
 - Midterm (0.5%)
 - Final (0.5%)

Ethics & Notice

- Sharing materials from the class with other individuals not in the class without written instructor permission will be treated as an Honor Code violation;
- Posting solutions and codes on **public sites** is also **prohibited**.

Programming Language

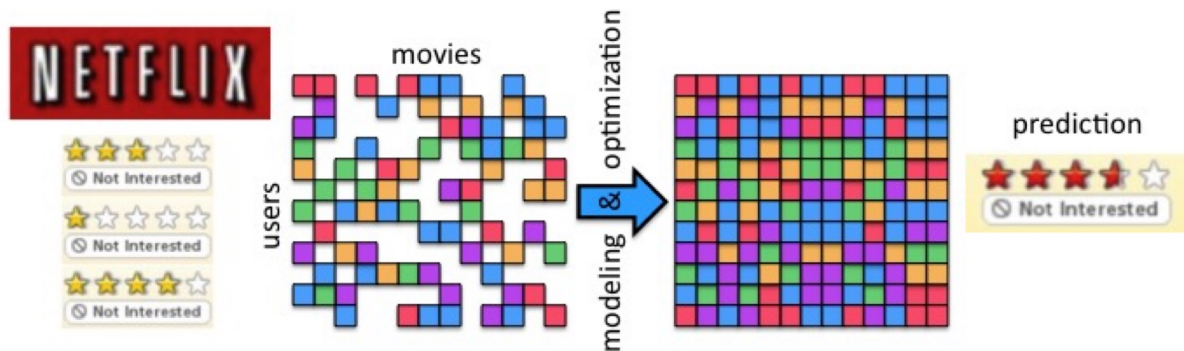
- We will mainly use **Python** for all coding exercises, and the GSI will teach a python review session



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Machine Learning is Everywhere



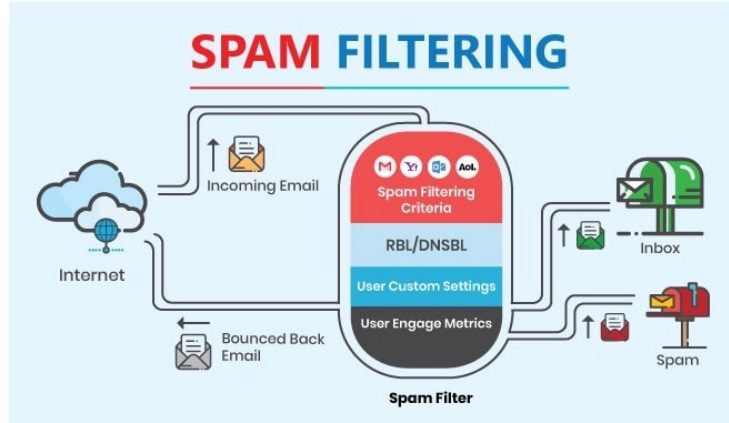
Recommender
System



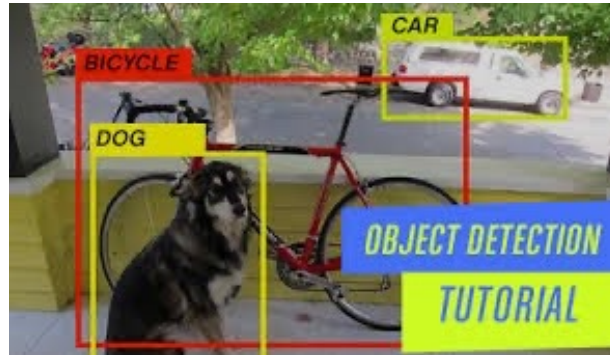
Machine
Translation

Machine Learning is Everywhere

Spam Filtering

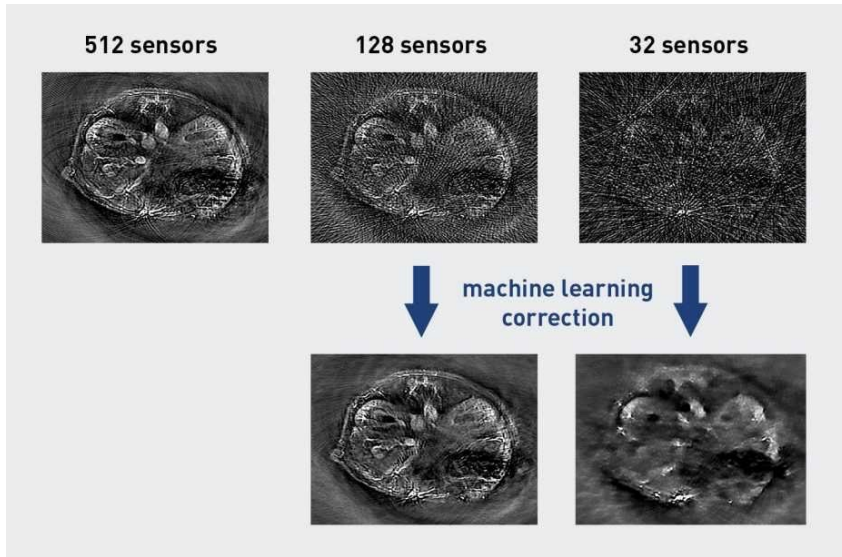


Object Recognition



Speech Recognition

Machine Learning is Everywhere



Medical Imaging

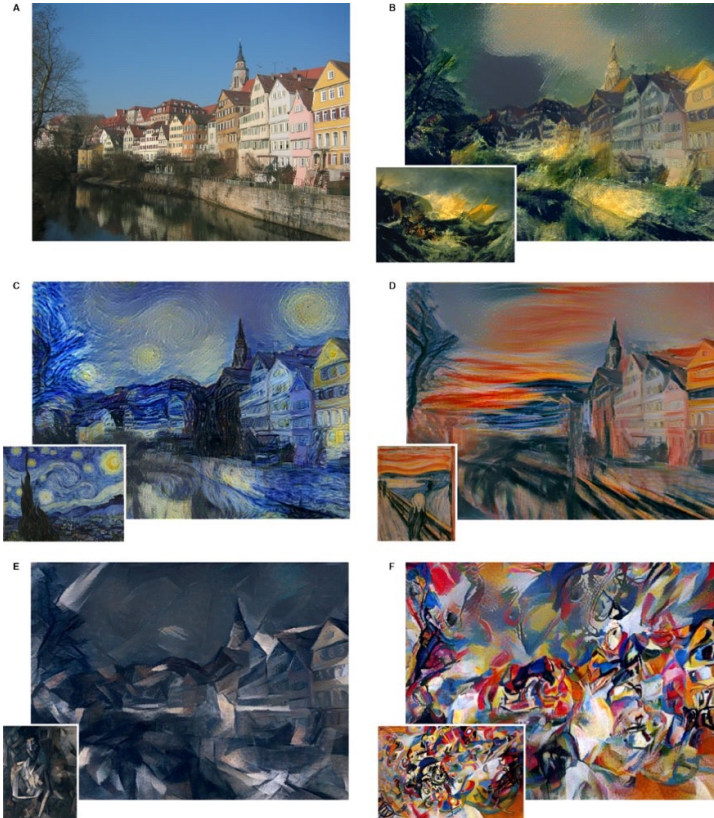


Autonomous driving



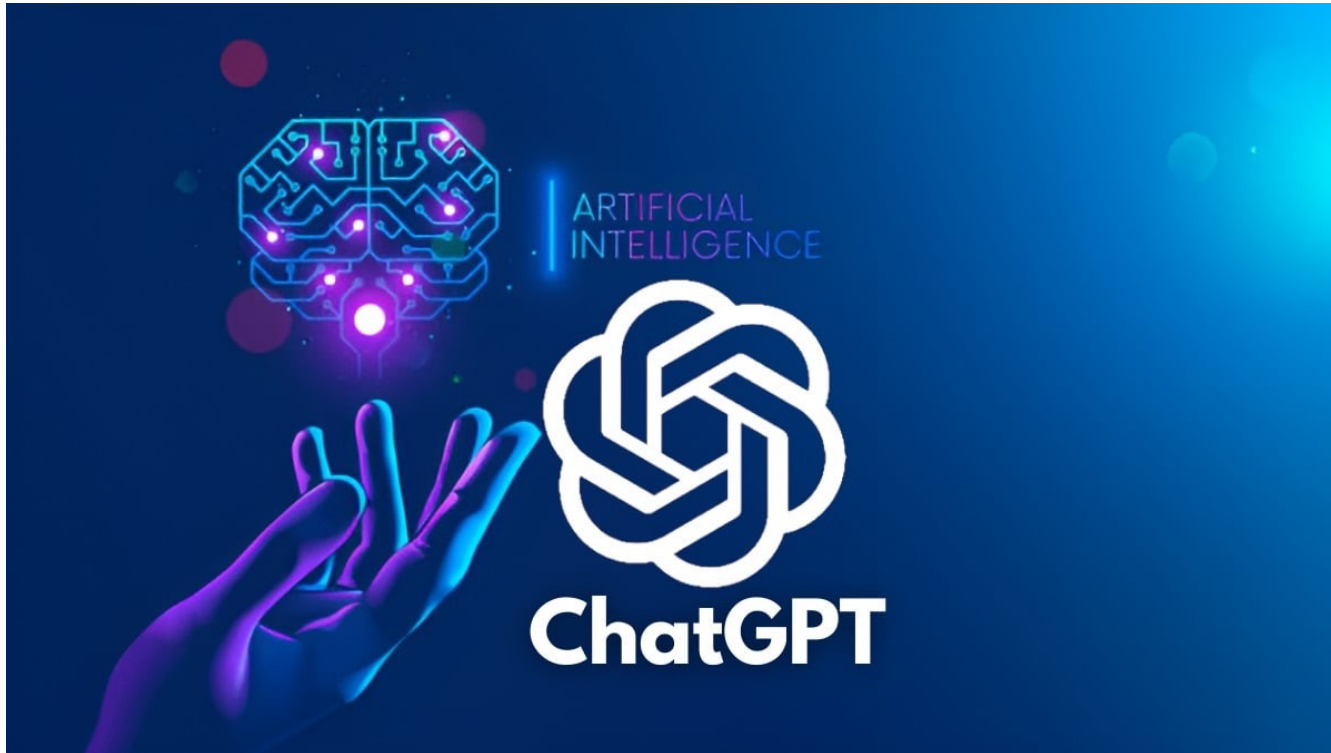
Healthcare

Machine Learning is Everywhere

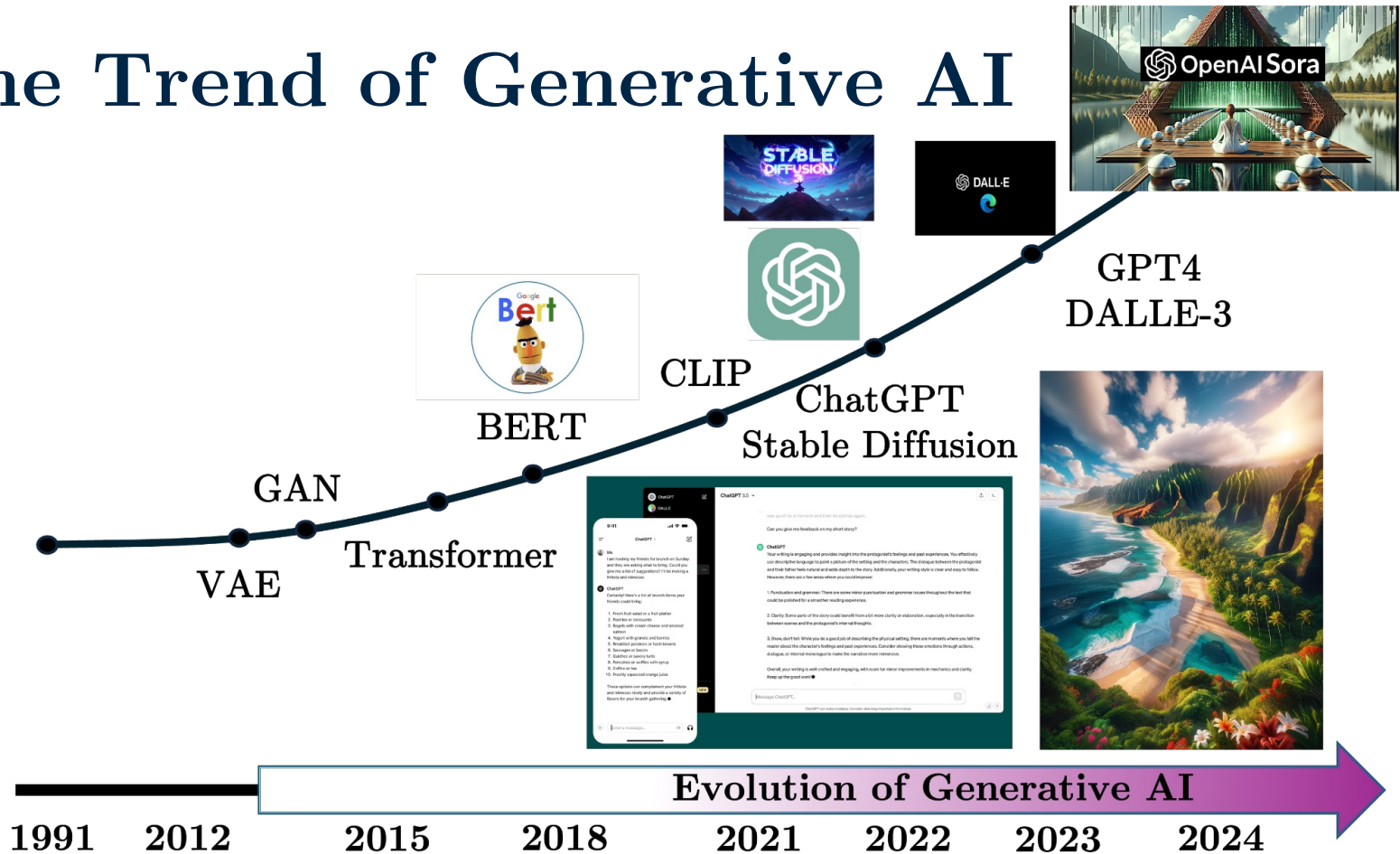


“Artificial artists”

Machine Learning is Everywhere



The Trend of Generative AI



What is Machine Learning?



- How can we
 - learn effective models from data?
 - apply these models to practical inference and all kinds of engineering problems (e.g., classification, clustering)?

What is Machine Learning?

To extract important patterns from data, understand “what the data says”, and use it to make predictions or decisions.

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- Extremely useful when:
 - *automate* something a human can do: face recognition, image classification, autonomous driving, etc.
 - do things a human *cannot* do: when the data is large-scale.

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Machine learning methods:
the mixture of statistics and optimization

Machine Learning & AI

Artificial Intelligence (AI): the high-level or cognitive capability of humans to reason and to think.

Machine Learning & AI

Artificial Intelligence (AI): the high-level or cognitive capability of humans to reason and to think.

Artificial Intelligence (AI) is the theory and development of computer systems able to **perform tasks normally require human intelligence.**

“Artificial Intelligence -- The Revolution Has Not Happened Yet”, -- Prof. Michael I. Jordan, UC Berkeley

Machine Learning & AI

Artificial Intelligence (AI): the high-level or cognitive capability of humans to reason and to think.

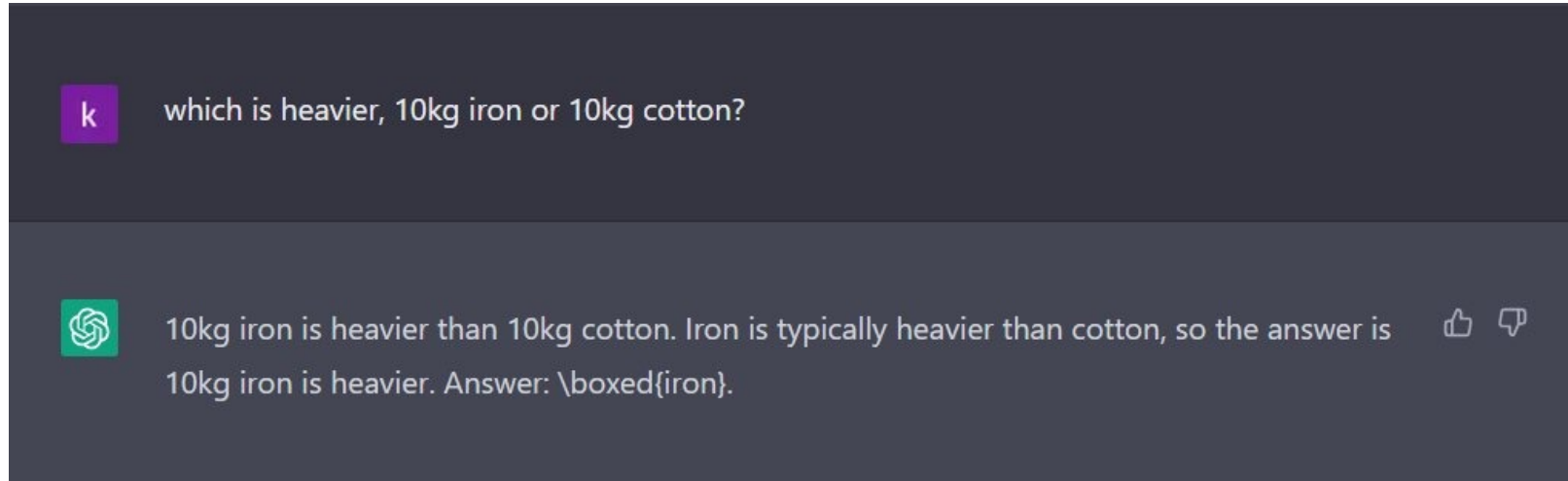


which is heavier, 10kg iron or 10kg cotton?

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Machine Learning & AI

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Deep learning, machine learning, and AI

Most of what is labeled AI today, particularly in the public sphere, is actually *machine learning*.

Artificial Intelligence

Is the field of study

Machine Learning

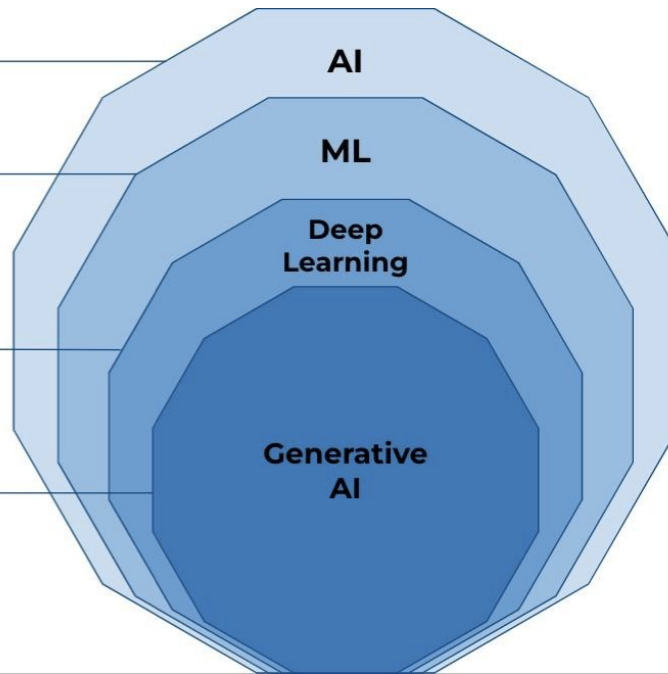
Is a branch of AI that focus on the creation of intelligent machines that learn from data.
Another very well know branch inside AI is **Optimization**.

Deep Learning

Is a subset of Machine Learning methods, based on **Artificial Neural Networks**.
Examples: CNNs, RNNs

Generative AI

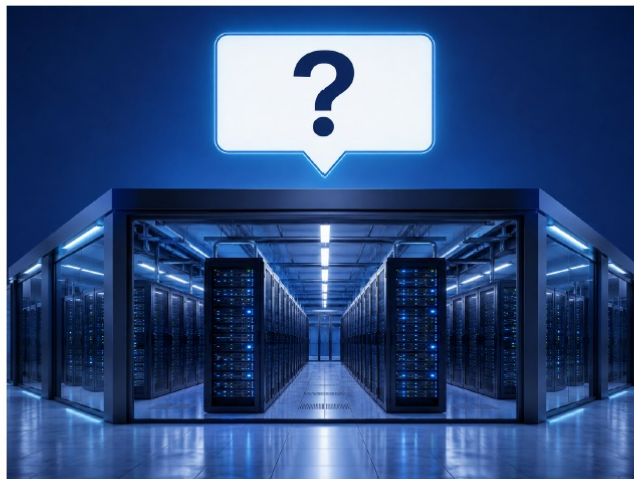
A type of ANNs that generate data that is similar to the data it was trained on.
Examples: GANs, LLMs



EECS 598: Foundations of Generative AI

Winter 2027, University of Michigan

Instructor: Clay Scott

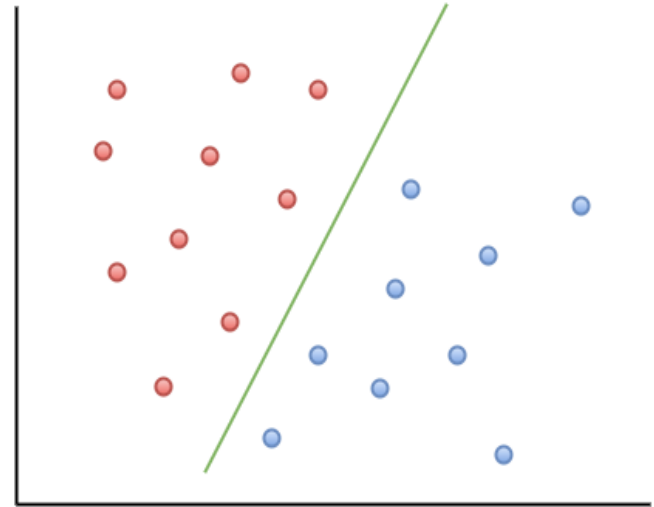


What algorithms run inside an AI data center?

Description: This course covers the algorithmic and mathematical foundations of modern generative language and image models: their architectures, training and inference procedures, and the principles governing their design. At least half of the course will be devoted to large language models, including autoregressive modeling and next-token prediction, self-attention layers and the transformer architecture, pre-training, post-training (including instruction fine-tuning, preference optimization, and RLHF), in-context learning, chain-of-thought prompting, reasoning models, test-time scaling, and alternative architectures. The remainder of the course will focus on diffusion and flow-matching models for image generation. Topics will include denoising and score-based formulations of diffusion models, flow matching, conditioning and guidance, and methods for training and sampling from these models.

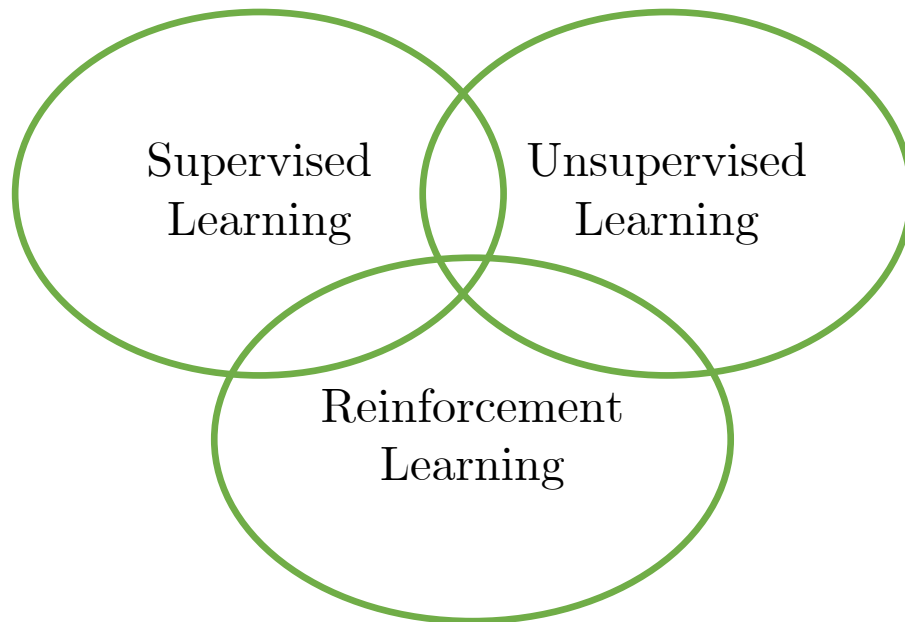
Ingredients of Machine Learning

- **Data:** from which we learn the patterns and trends
- **Hypotheses/Models:** devised to capture the patterns in data
- **Learning:** design algorithms to learn the model from the data
- **Prediction:** apply the learned model to predict the future outcomes.



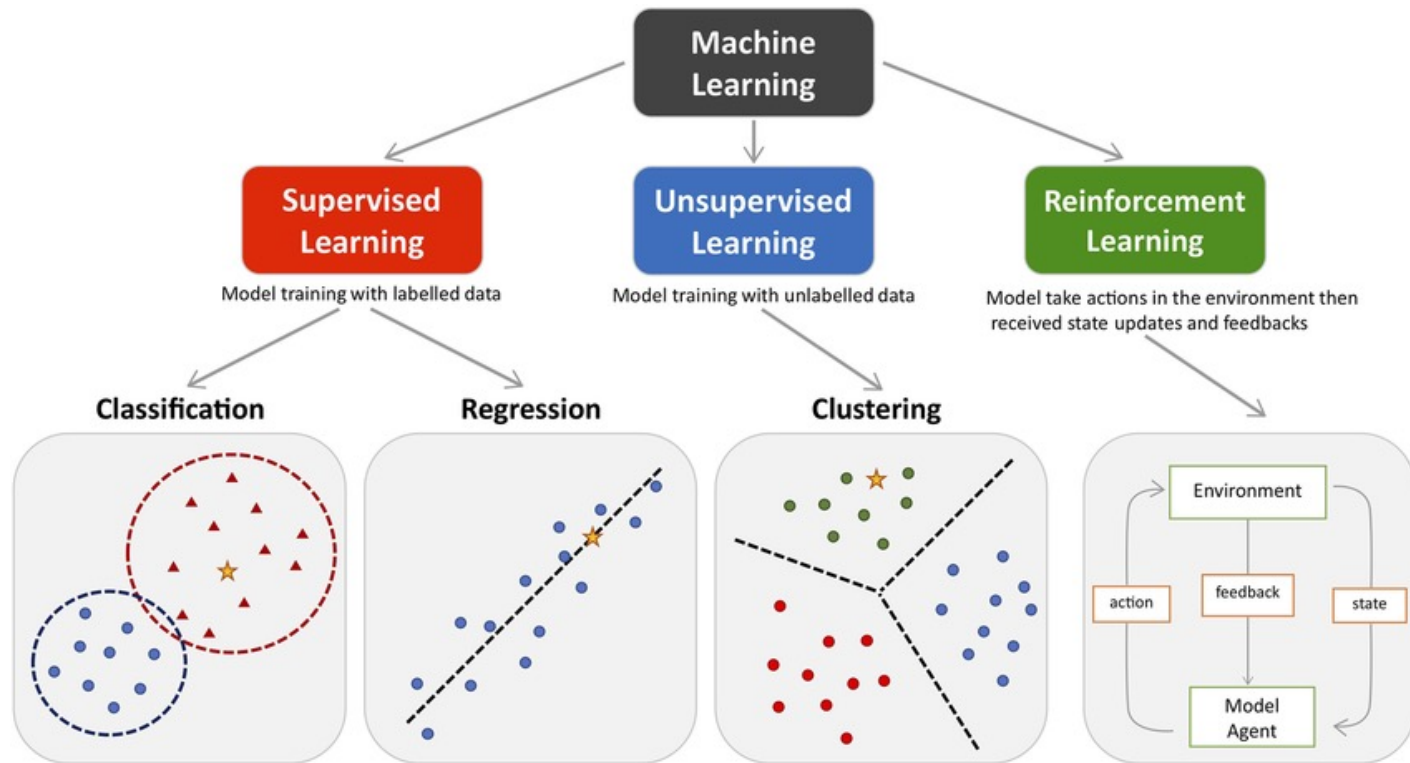
Taxonomy of Machine Learning

(A Simplistic View Based on Tasks)



Can also be viewed as tools/methods

Taxonomy of Machine Learning



Settings of Machine Learning

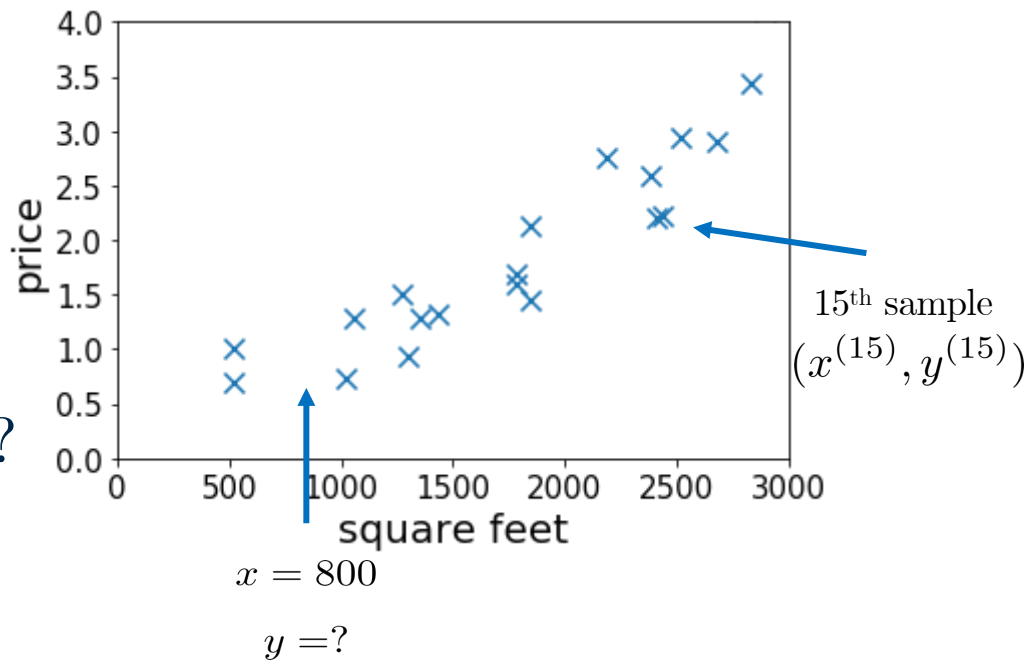
- **Supervised Learning**
 - We are given data and **labels** for those data
 - e.g., pictures of dogs and cats with the label dog, cat
 - e.g., credit card data with fraud events flagged
- **Unsupervised Learning**
- **Reinforcement Learning**

Example: Housing Price Prediction

- **Given:** a dataset that contains n samples

$$(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)})$$

- **Task:** if a residence has x square feet, predict its price?

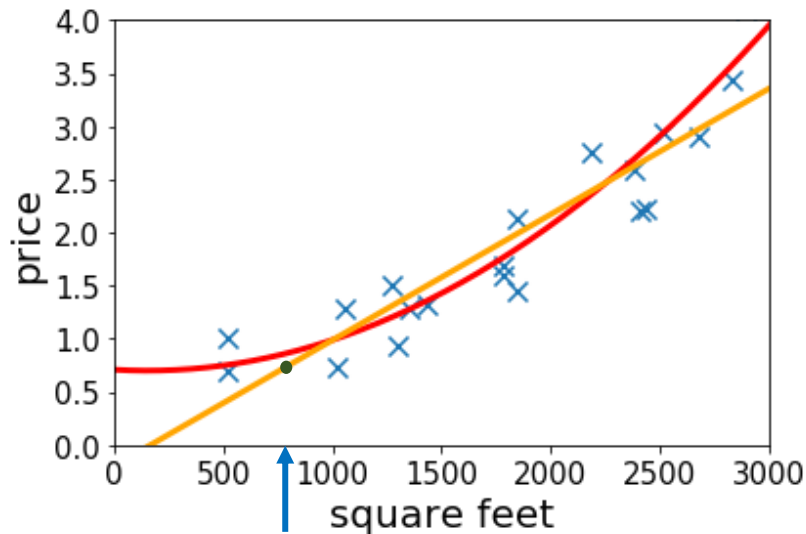


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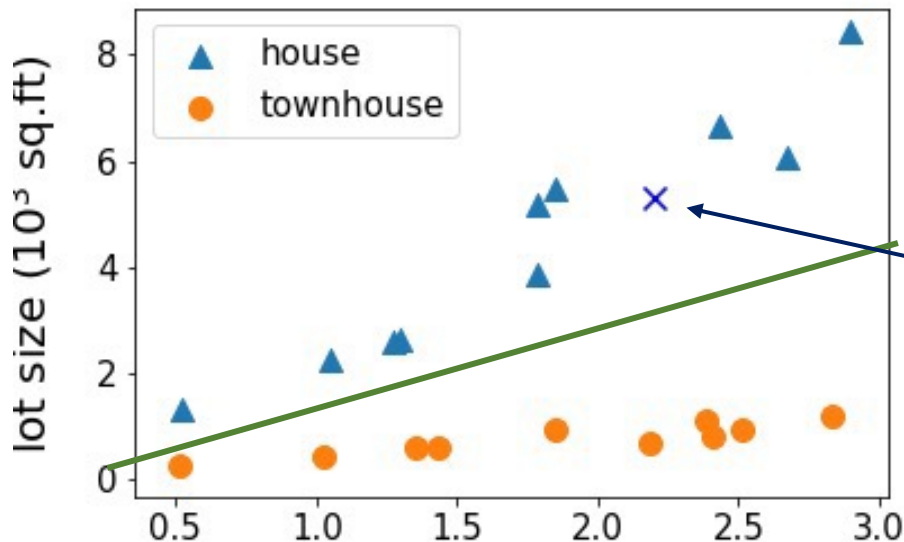


$x = 800$

$y = ?$

Regression vs Classification

(size, lot size) $\rightarrow$ house or townhouse?



$y = \text{house or townhouse?}$

Regression vs Classification

- **Regression:** if $y \in \mathbb{R}$ is a **continuous variable**
e.g., price prediction
- **Classification:** the label $y \in \mathbb{R}$ is a **discrete variable**
e.g., the task of predicting the types of residence

Supervised Learning in Computer Vision

- Image Classification

x = raw pixels of the image, y = the main object

ILSVRC

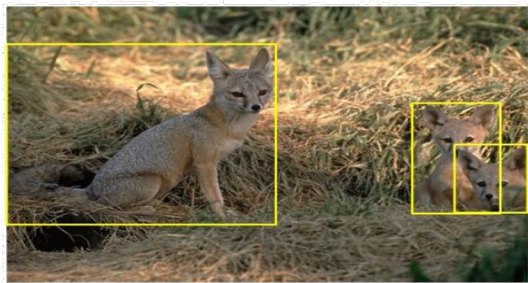


ImageNet Large Scale Visual Recognition Challenge. Russakovsky et al.'2015

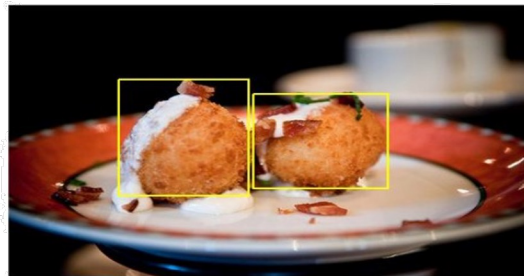
Supervised Learning in Computer Vision

- Object localization and detection

x = raw pixels of the image, y = the bounding boxes



kit fox



croquette



airplane

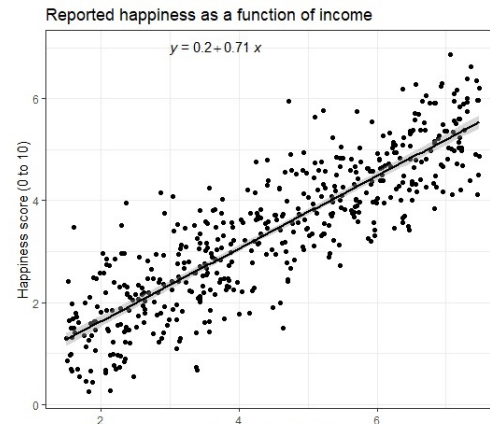


frog

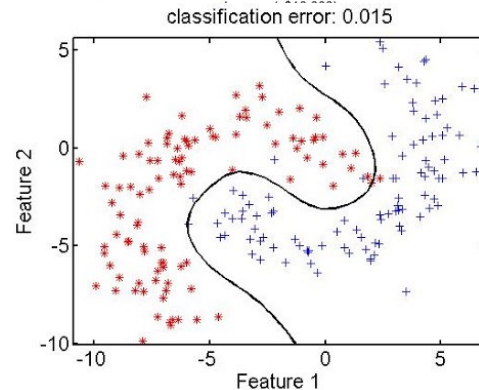
ImageNet Large Scale Visual Recognition Challenge. Russakovsky et al.'2015

Settings of Machine Learning

- **Supervised Learning**
 - How are the data x related to the label y ?
 - How are the data and label (x, y) related to unseen new data?
- Unsupervised Learning
- Reinforcement Learning



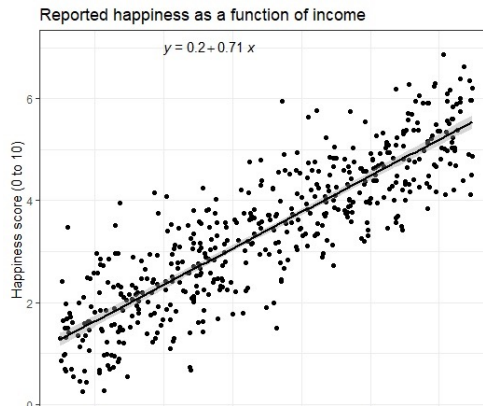
Linear
Models



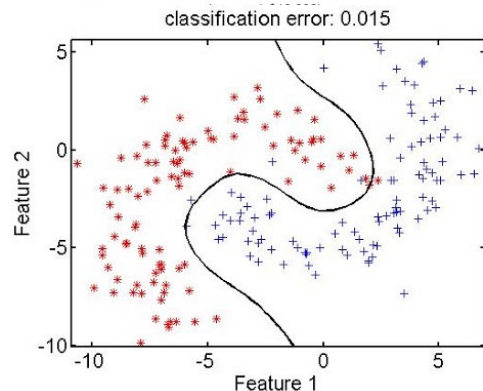
Nonlinear
Models

Settings of Machine Learning

- **Supervised Learning**
 - Linear and logistic regression
 - Support vector machines
 - Neural networks
 - Decision Trees
 - Naïve Bayes
- Unsupervised Learning
- Reinforcement Learning



Linear
Models



Nonlinear
Models

Settings of Machine Learning

- **Supervised Learning**
 - we are given **both data and labels** for those data
 - e.g., pictures of dogs and cats with the label dog, cat
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Settings of Machine Learning

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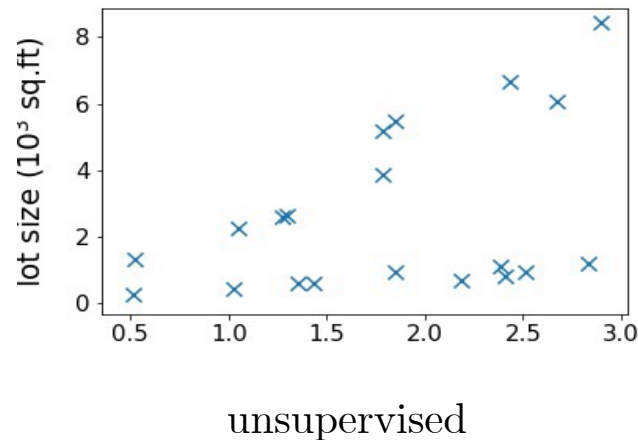
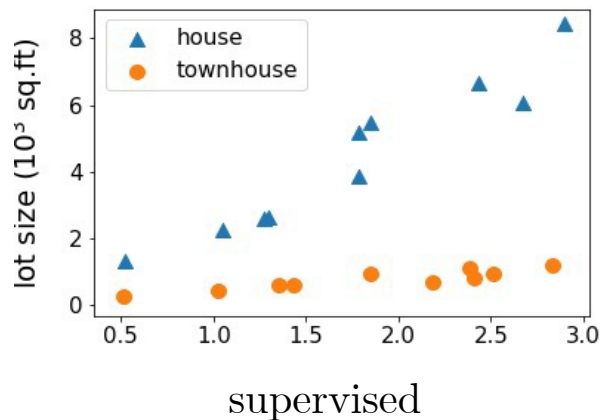
- **Unsupervised Learning**

- we are given **only data**.
- e.g., a bunch of animal pictures with no label
- e.g., credit card data **without any tags**

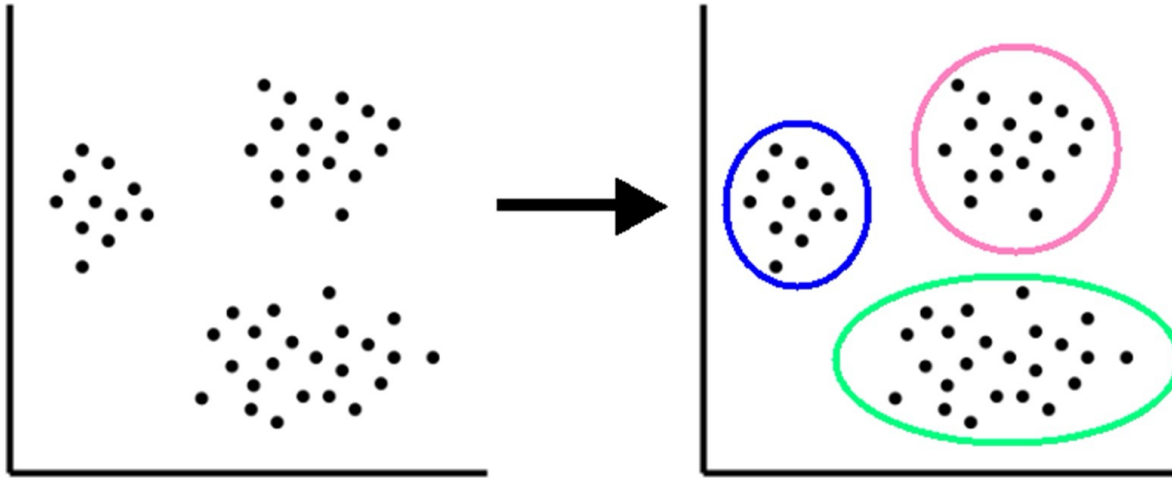
- **Reinforcement Learning**

Unsupervised Learning

- Dataset contains **no labels**: $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}$
- **Goal (vaguely-posed)**: to learn underlying structures in the data

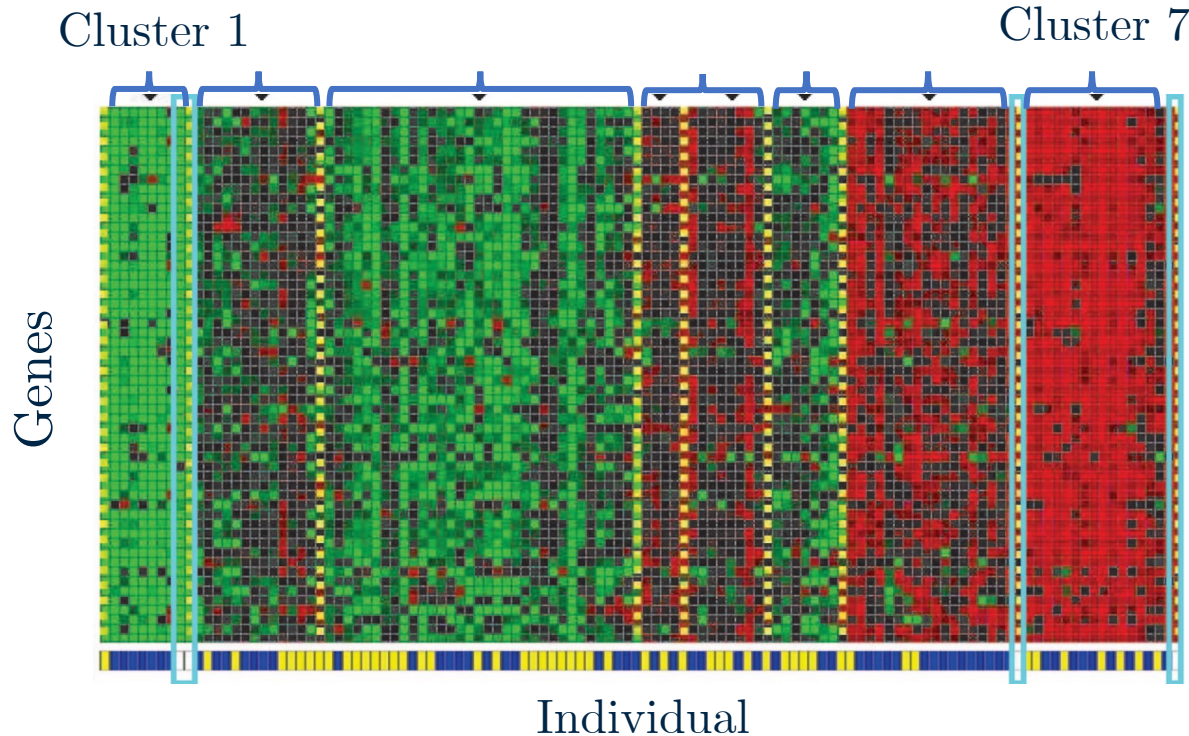


Clustering



Clustering

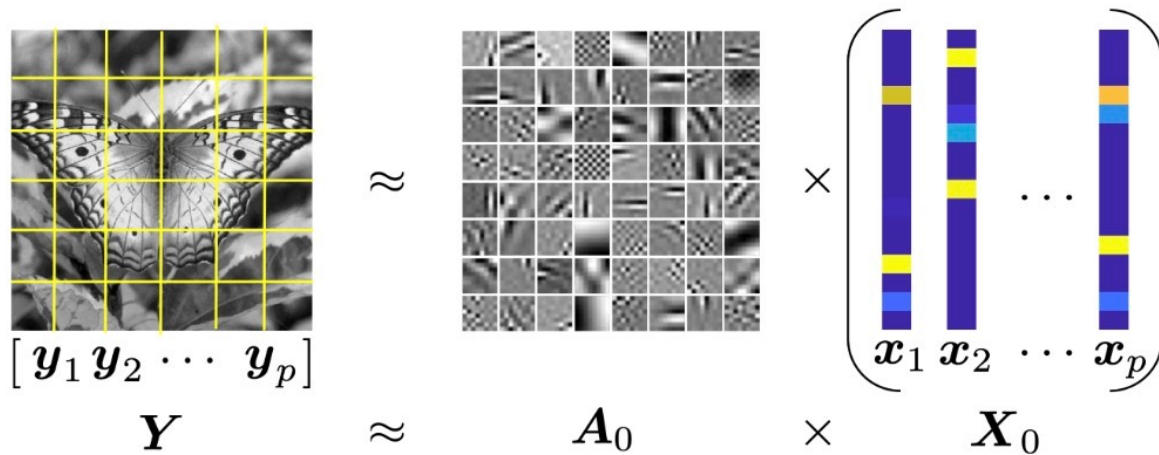
Example: Clustering Genes



Identifying Regulatory Mechanisms using Individual Variation Reveals Key Role for Chromatin Modification. [Su-In Lee, Dana Pe'er, Aimee M. Dudley, George M. Church and Daphne Koller. '06]

Representation Learning

- Example: Dictionary learning



$$\min_{A \in \mathcal{M}, X} \underbrace{f(Y, A \cdot X)}_{\text{data fidelity}} + \lambda \cdot \underbrace{g(X)}_{\text{regularizer}}$$

Representation Learning



Denoising

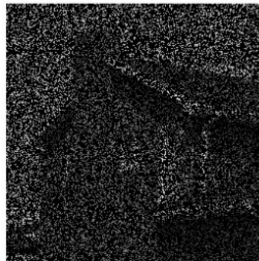


Image Restoration



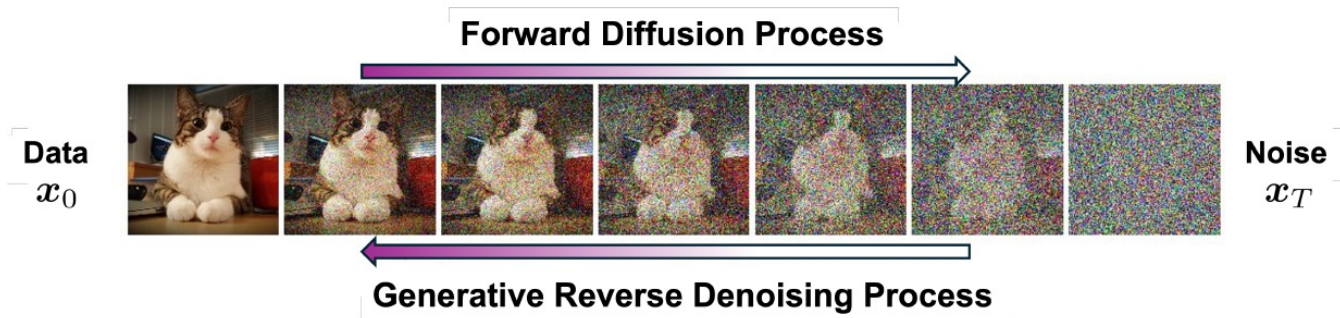
Super Resolution



Image Half-toning

- Image courtesy of Julien Mairal et al.

Generative Models



- **Forward process:** progressively adding noise to an image x_0 ;²

$$x_t = \alpha_t x_0 + \beta_t \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, I).$$

- **Backward process:** starting from a random noise ϵ , progressively denoising to generate an image x_0 .

Generative Models



Text-to-Image Generation
(Stable Diffusion 3.5, Stability AI)

Autoregression: ✓ High quality ✓ Arbitrary-length ✓ KV caching ✗ Not Parallelizable

Generation steps

There are three categories of the average rate
There are three categories of the average rate
There are three categories of the average rate of...

Diffusion: ✗ Lower quality ✗ Fixed-length ✗ No KV caching ✓ Parallelizable

the resusability will continue to the
Repeal the resusability cuts and the law will continue to reduce the
Repeal the resusability cuts and prove the law will continue to reduce the deficit.

Block Diffusion: ✓ High quality ✓ Arbitrary-length ✓ KV caching ✓ Parallelizable

On September 17, we will be giving the release of
On September 17, 2016, we will be giving the release of
On September 17, 2016, we will be giving the beta-release of the to our server testing ...

Language Generation
(Discrete Diffusion, Arriola et al.'25)



Video Generation
(VEO, Google)

Settings of Machine Learning

- **Supervised Learning**

- How are the data $\mathbf{x}$ related to the labels y ?
- How are the data and labels $(\mathbf{x}, y)$ related to unseen new data?

- **Unsupervised Learning**

- What kinds of structures are in the data $\mathbf{x}$?
- How is the structure in the data $\mathbf{x}$ related to structure in new unseen data?

- **Reinforcement Learning**

Settings of Machine Learning

- Supervised Learning
- **Unsupervised Learning**
 - Clustering
 - Dimension reduction
 - Representation learning
 - Probability distribution estimation
- Reinforcement Learning

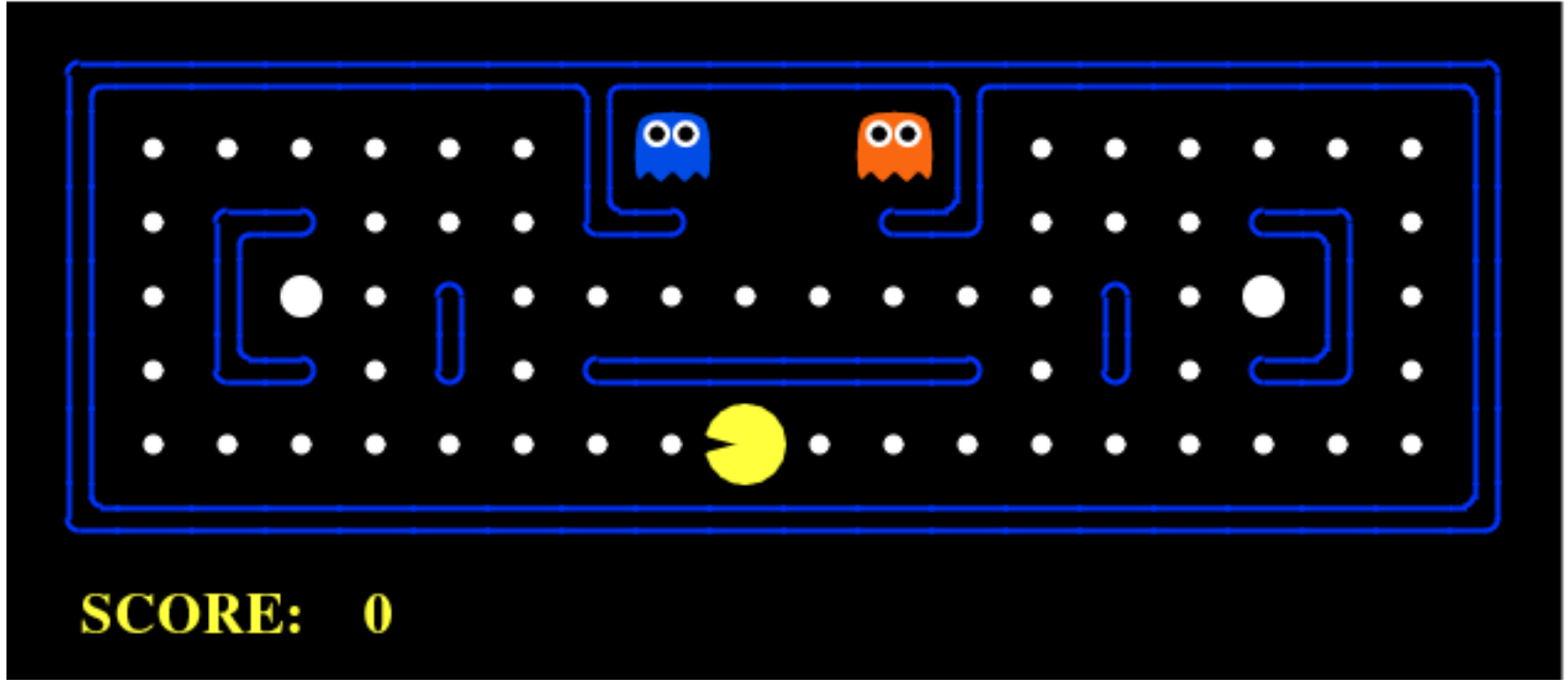
Settings of Machine Learning

- Supervised Learning
- Unsupervised Learning
- **Reinforcement Learning**
 - Training of machine learning models to make a sequence of decisions interactively
 - Learns to achieve a goal in an uncertain, potentially complex environment

Reinforcement Learning

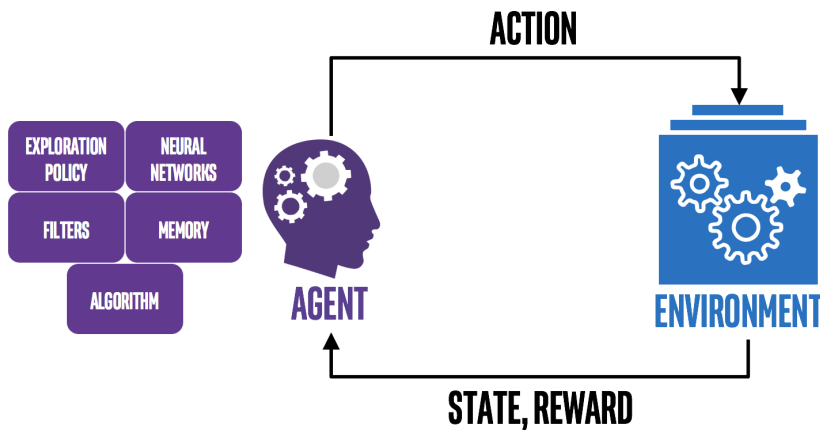


Reinforcement Learning



Reinforcement Learning

The intelligent agent learns to take actions in an environment interactively to *maximize cumulative reward*.



Try the strategy
and collect
feedback

Improve the
strategy based on
the feedback

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Course Goals

The course goal is to grasp *mathematical principles*, *foundational algorithms* of machine learning, through basic-theoretical analysis and implementations

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- **Supervised learning** (e.g., regression, classification, neural networks & deep learning, etc.);
- **Unsupervised learning** (e.g., clustering, mixture models, dimension reduction, etc.);

Contents & Syllabus

- **Supervised Learning (9 weeks)**
 - Regression and linear prediction (linear models)
 - Support vector machines (linear models)
 - Kernel methods (nonlinear models)
 - Deep neural networks (nonlinear models)
- **Unsupervised Learning (5.5 weeks)**
 - Clustering (K-Means, Mixture of Gaussians, EM)
 - Dimension reduction: PCA, autoencoder
 - Representation learning

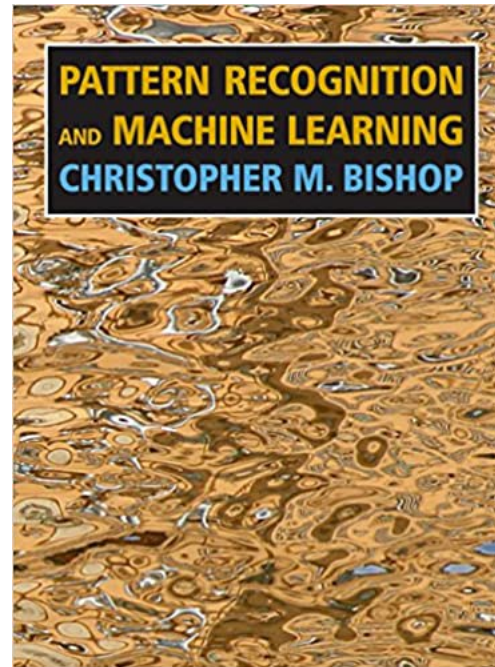
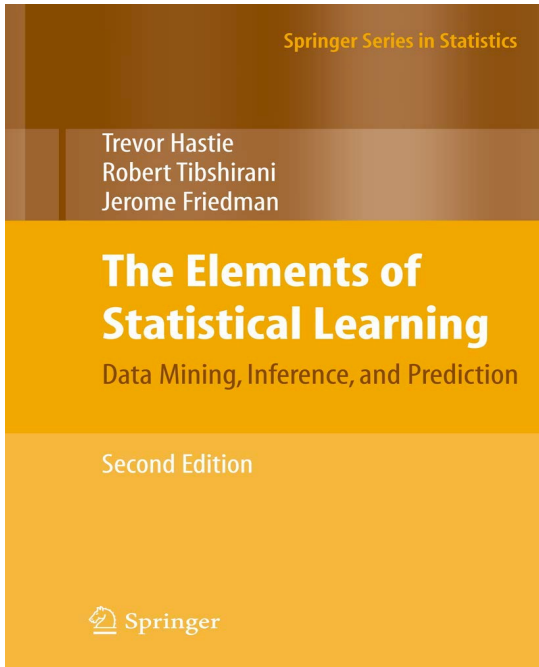
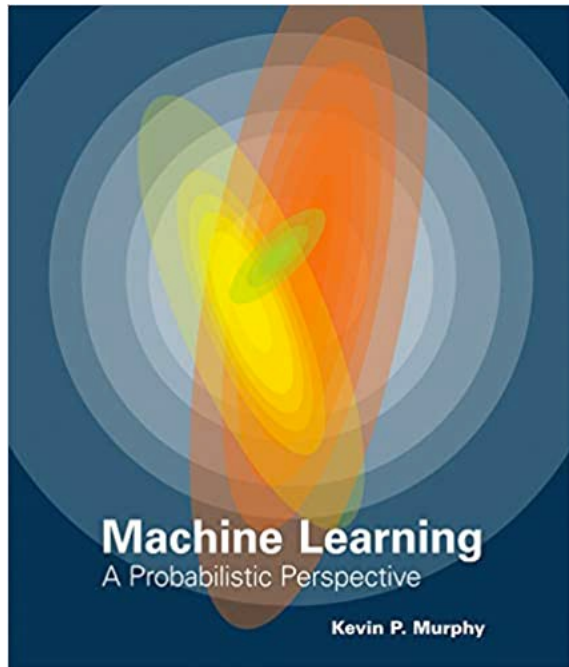
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- The Elements of Statistical Learning: Data mining, Inference, and prediction. Hastie, T., Tibshirani, R., & Friedman, J.
- Pattern Recognition and Machine Learning, Bishop, C. M. (2006), Springer.
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Books & References



Questions?